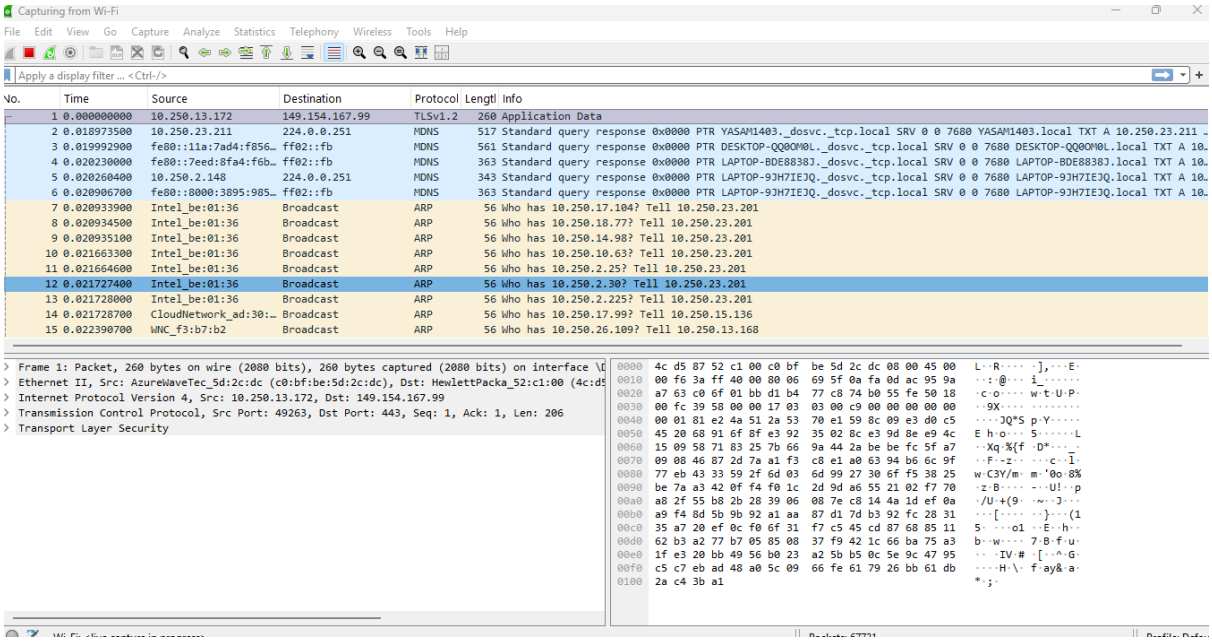
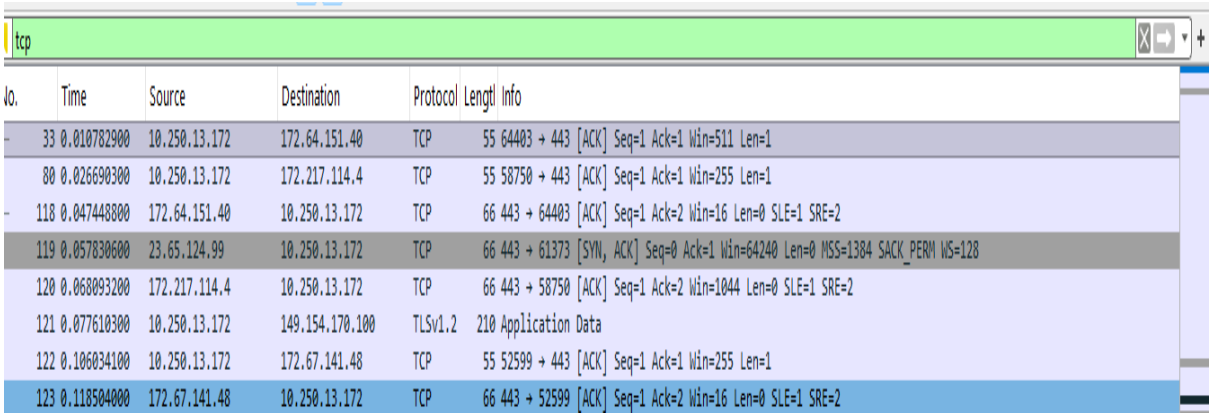
TASK1:

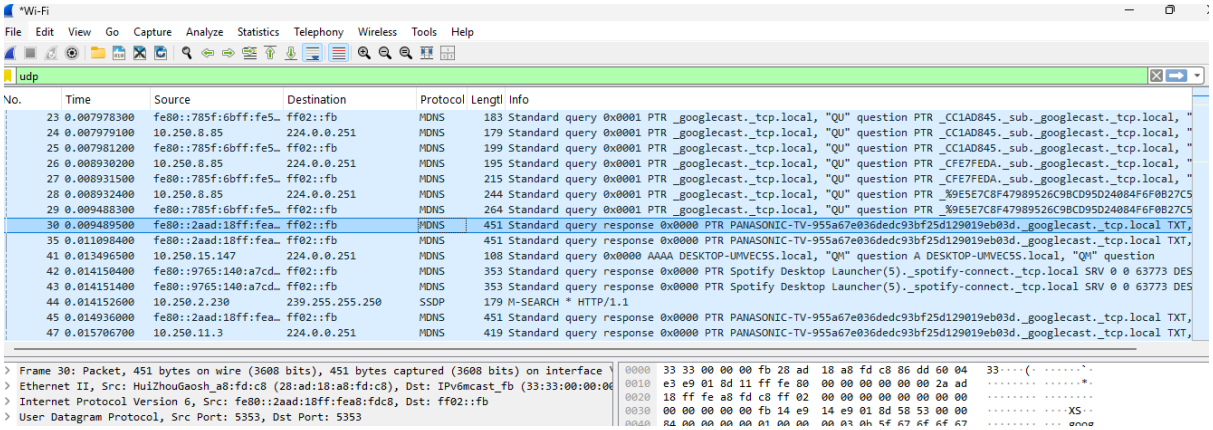
1.Open the PCAP file and identify the total number of packets captured.



2.Apply the filter tcp and record the number of TCP packets.

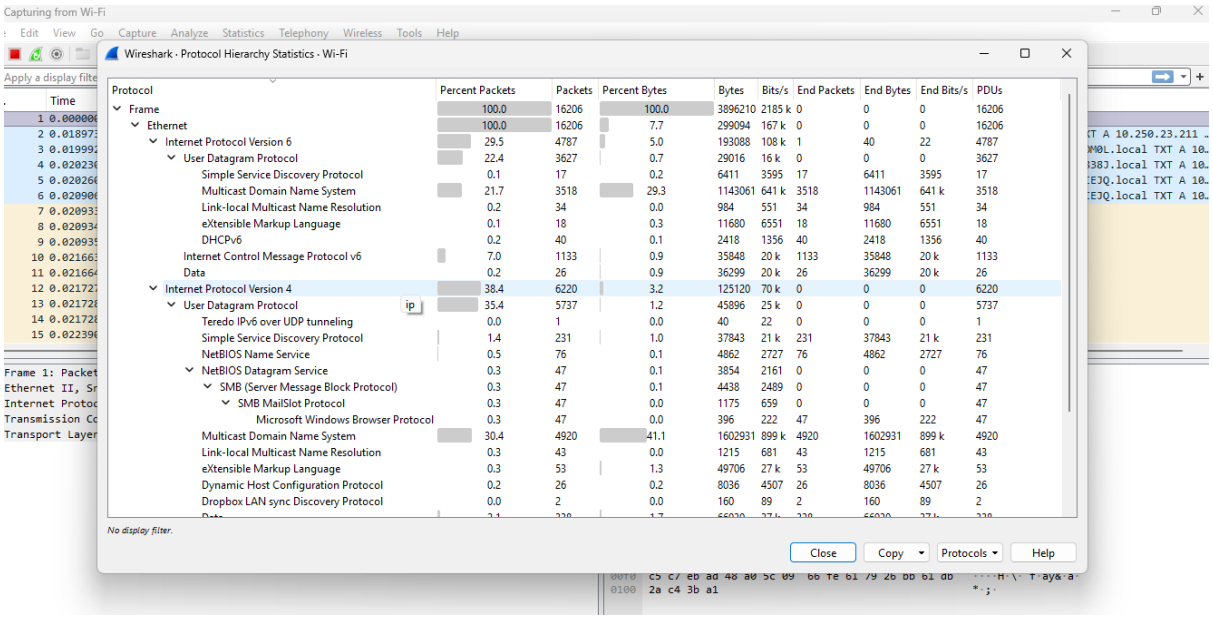


3.Apply the filter udp and record the number of UDP packets.



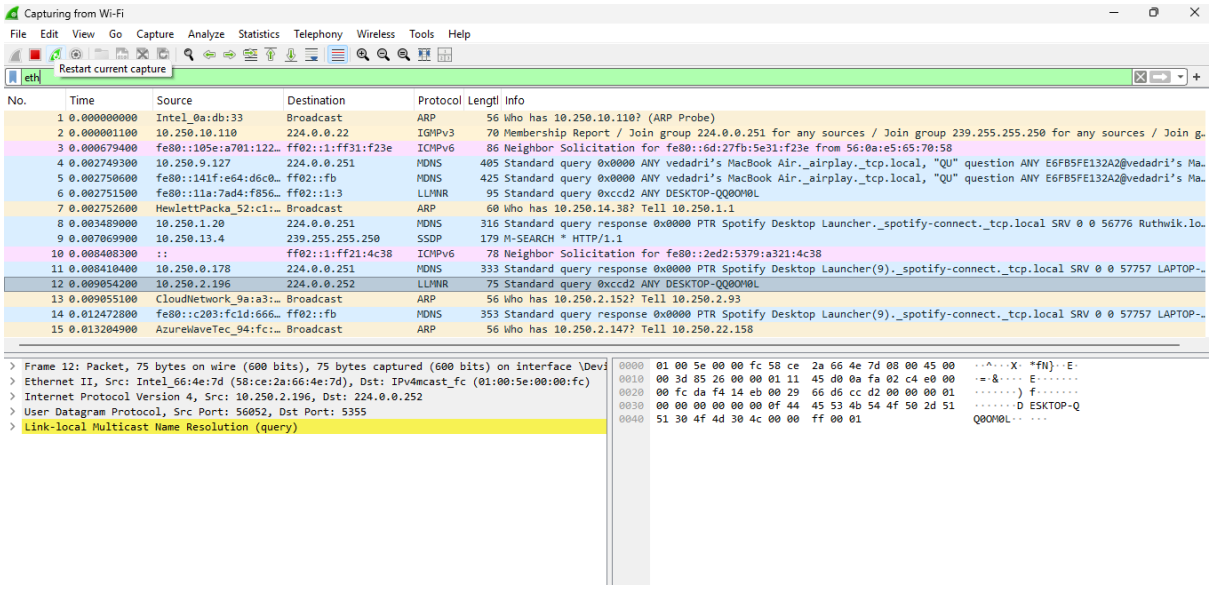
4.Apply the filter icmp and record the number of ICMP packets.

5.identify the top five protocols using Statistics → Protocol Hierarchy.



TASK2:

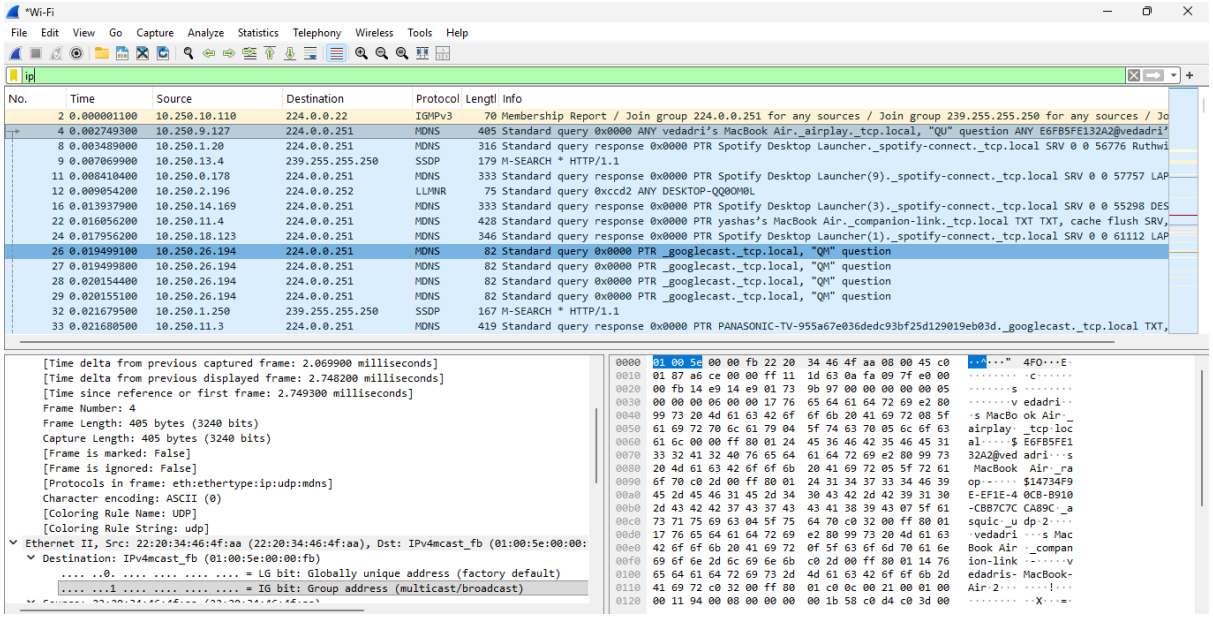
6.Apply the filter eth.



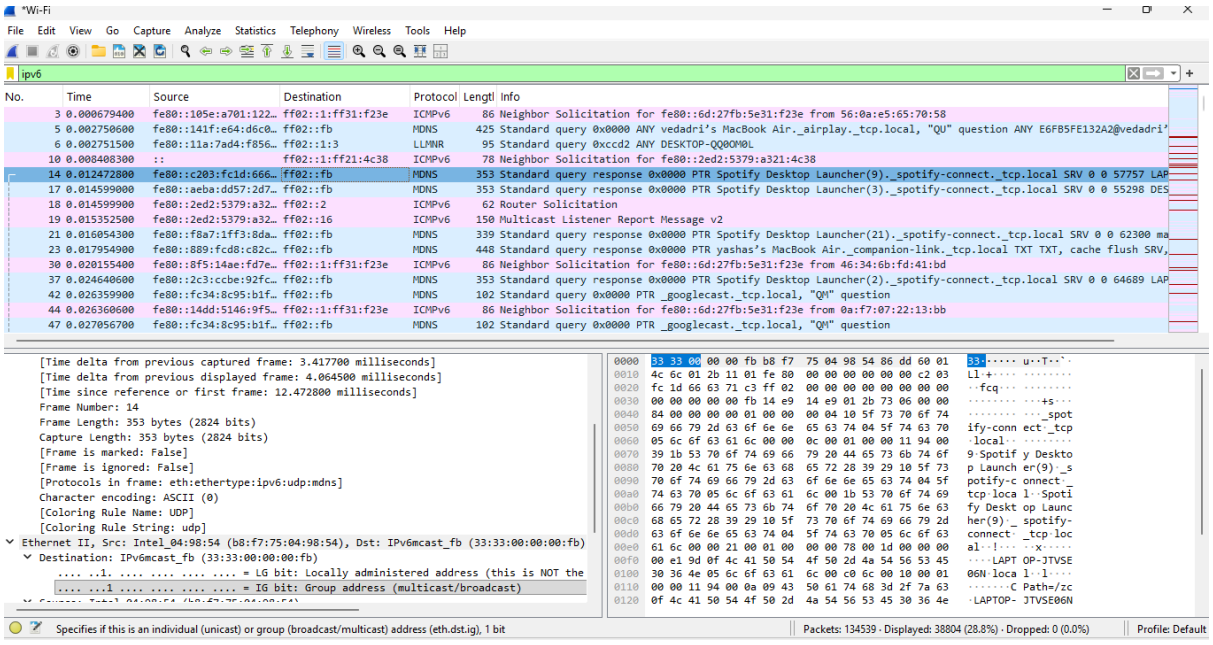
7. Select five different packets and record their:

Packet	Frame no	Frame length	Source MAC	Destination MAC	Ether Type
75	12	75	58:ce:2a:66:4e:7c	01:00:5e:00:00:fc	0x0800
60	7	60	4c:d5:87:52:c1:00	ff:ff:ff:ff:ff:ff	0x08006
1514	18	1514	00:1a:2b:3c:4d:5e	33:33:00:00:00:16	0x86dd
74	23	74	58:ce:2a:66:4e:7c	4c:d5:87:52:c1:00	0x0799
98	31	98	4c:d5:87:52:c1:00	58:ce:2a:66:4e:7c	0x0897

8. Identify packets carrying IPv4 and IPv6 traffic.

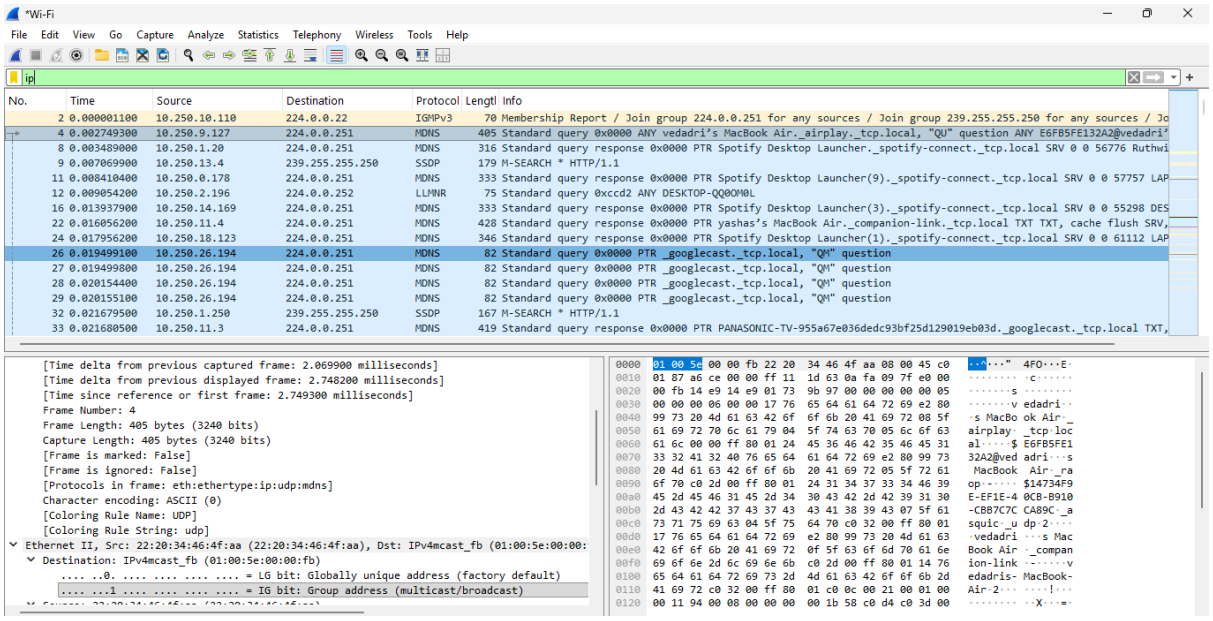


Ipv6



Task 3: IPv4 packet inspection

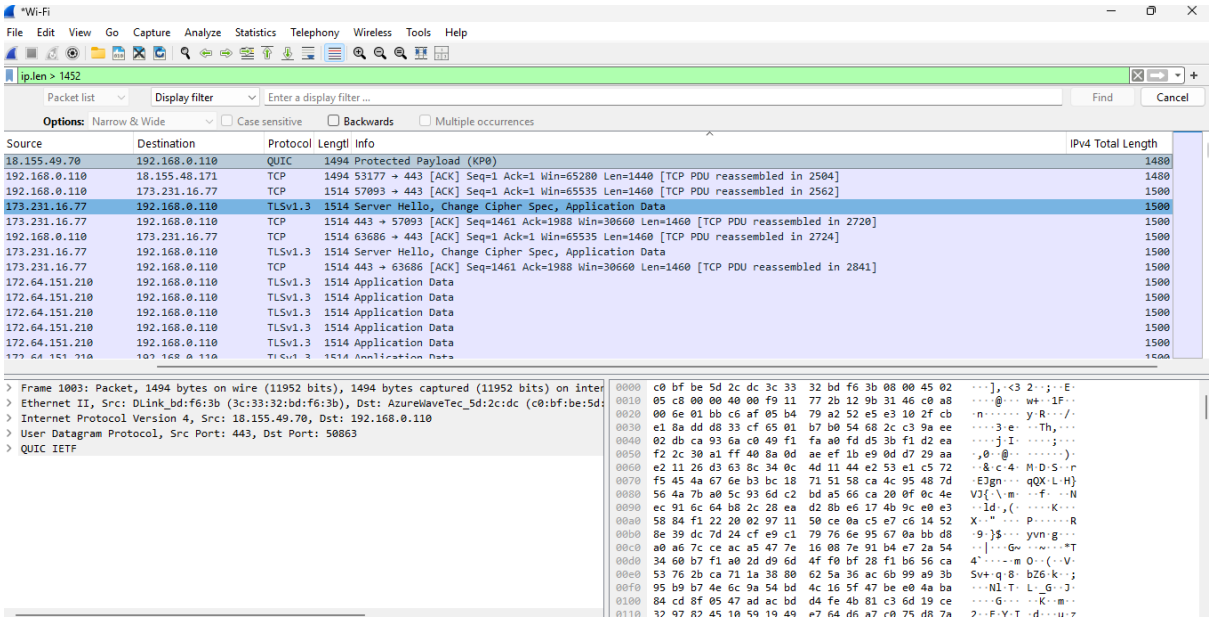
9. Apply the filter ip.



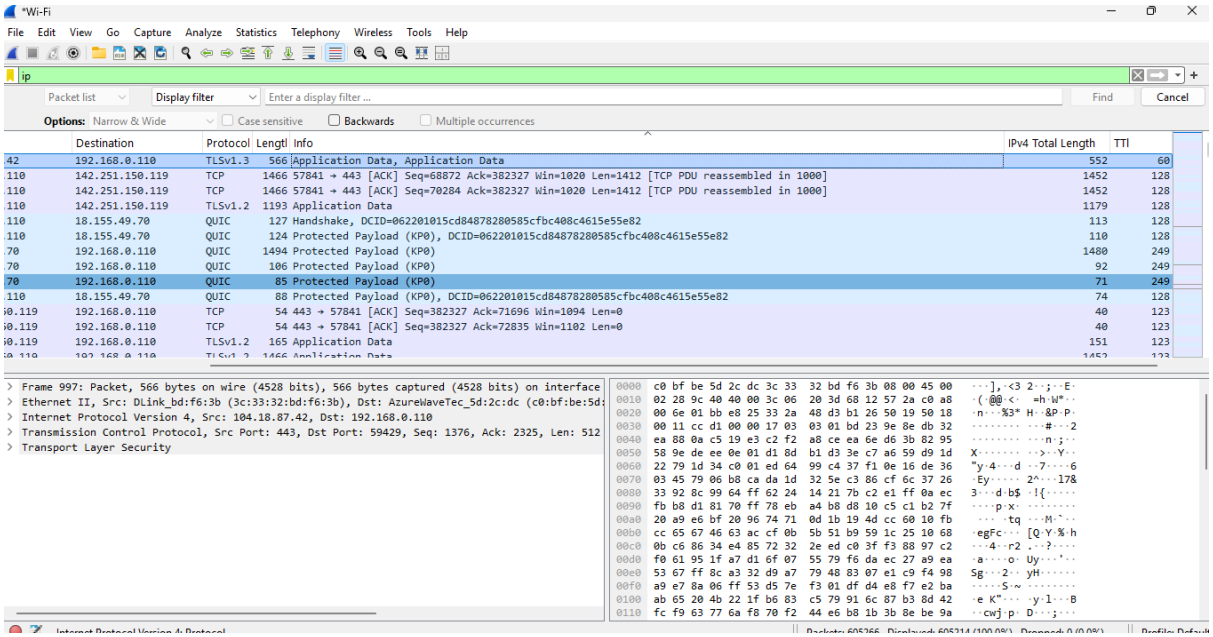
10. Identify five IPv4 packets and record:

Packet	Source IP Address	Destination IP Addres	HeaderLength	Tota Length	TTL	Identification
75	10.250.2.196	224.0.0.252	20	61	1	0x8526
82	10.250.26.194	224.0.0.251	20	68	1	0x7d02
33	10.250.11.3	224.0.0.251	20	405	1	0xb09a
41	10.250.29.142	224.0.0.251	20	68	1	0xec7e
45	192.168.1.25	224.0.0.252	24	120	1	0x9d3f

11. Find the packet with the maximum IPv4 total length.

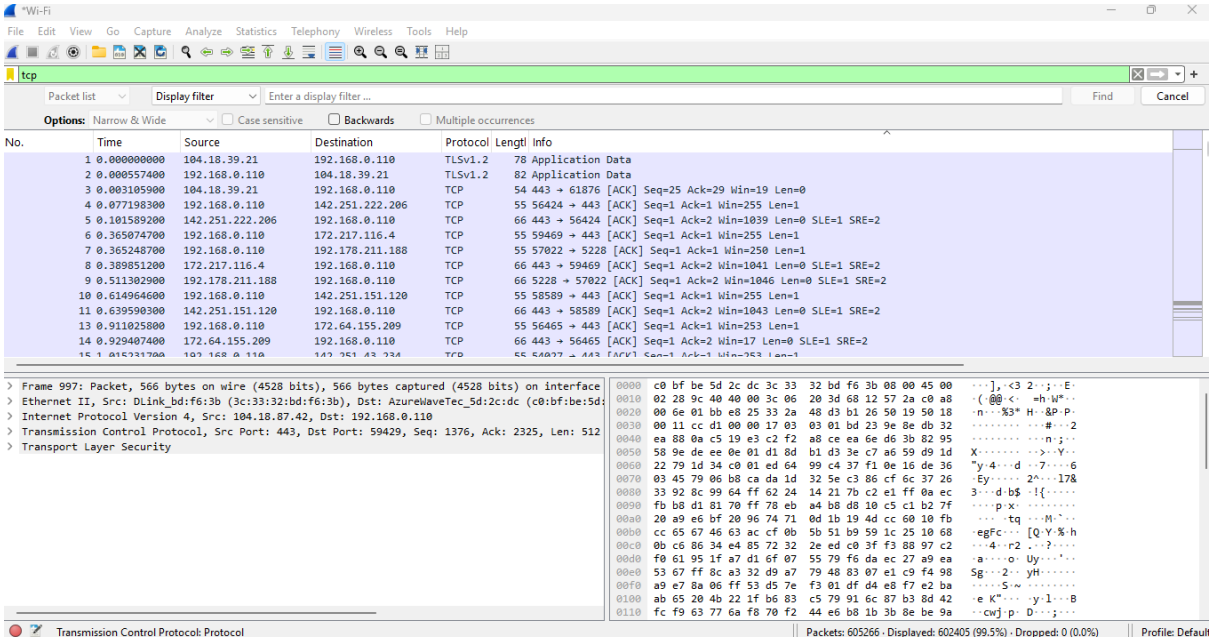


12. Identify packets with different TTL values.



Task 4: TCP header inspection

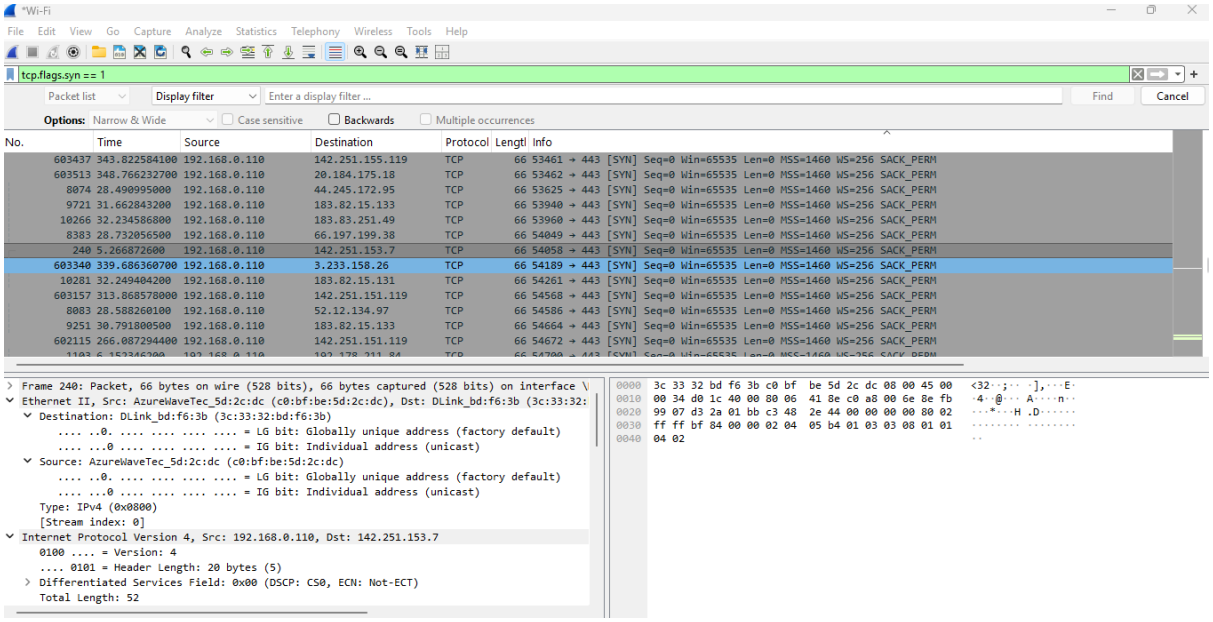
13. Apply the filter tcp.



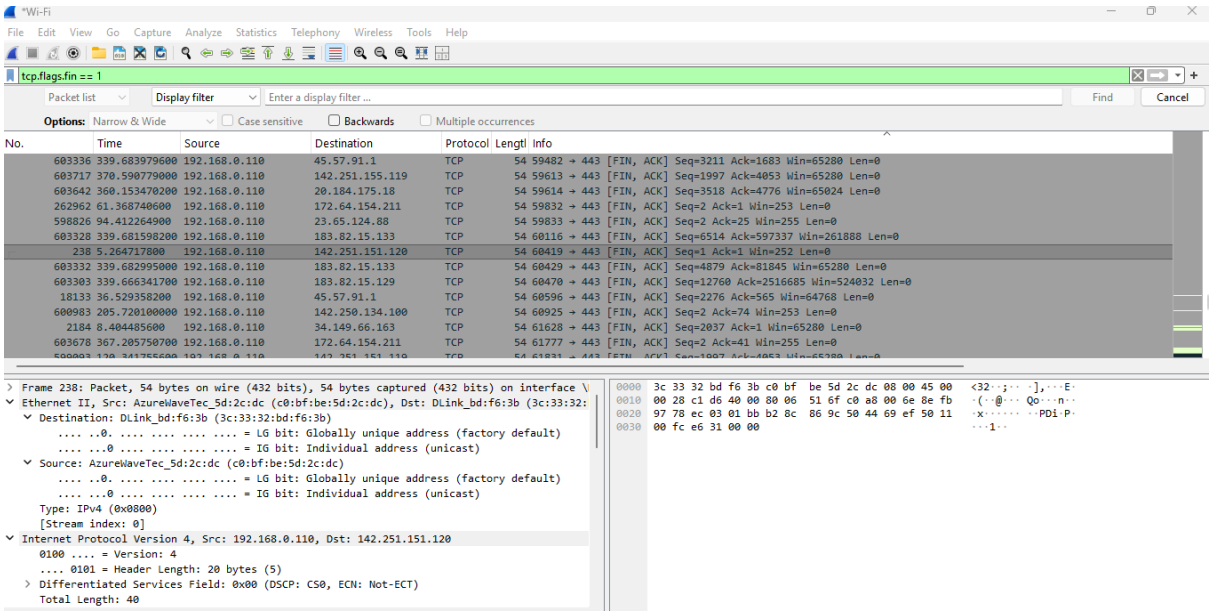
14. Select five TCP packets and record:

Source port	Destination port	Sequence no	Acknowledgement no	Tcp headerlength	payload
443	61876	1	2	20	24
61875	443	1	25	20	28
520	2200	1	32	20	27
6750	130	1	20	20	29
720	45000	1	43	20	32

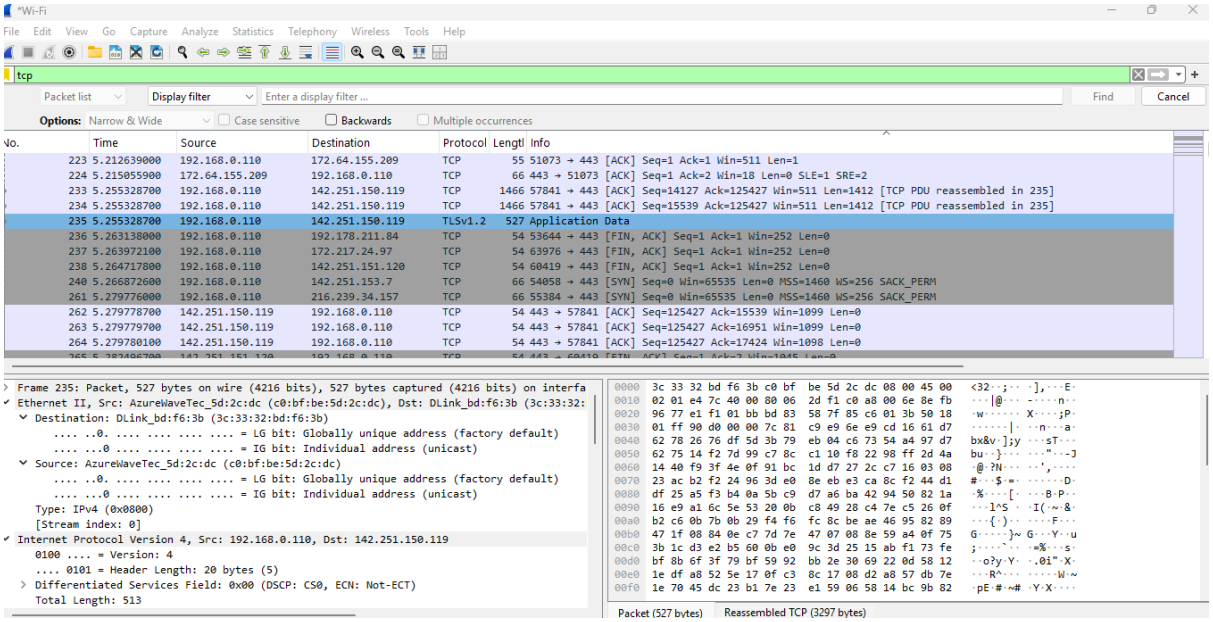
15.Apply the filter tcp.flags.syn == 1 and identify TCP connection establishment packets

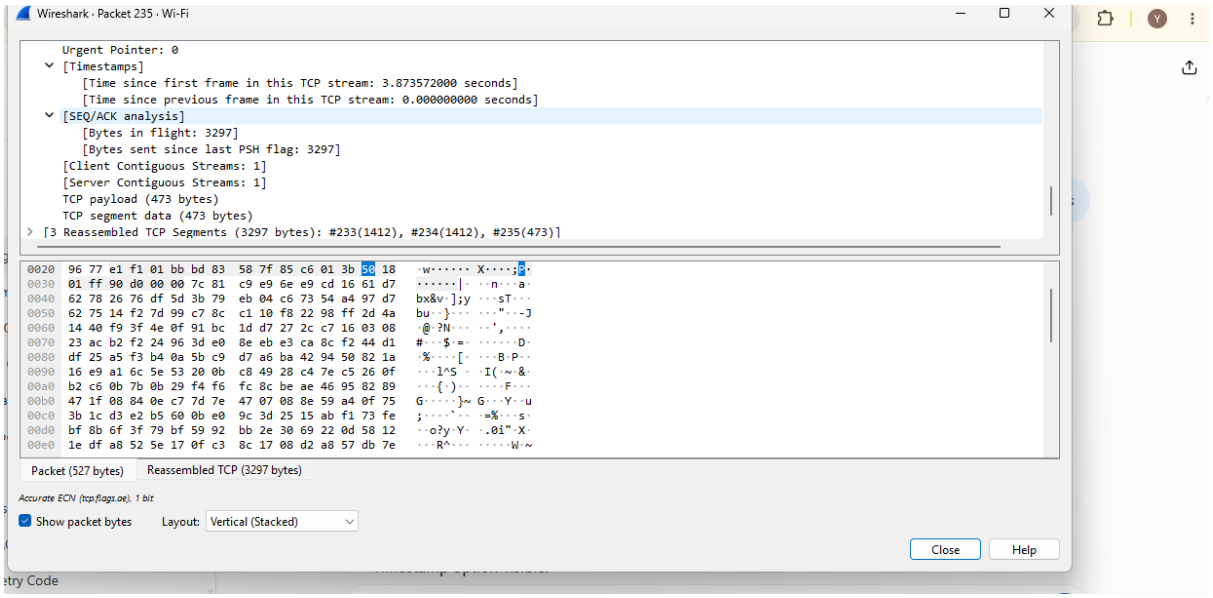


16. Apply the filter tcp.flags.fin == 1 and identify connection termination packets.

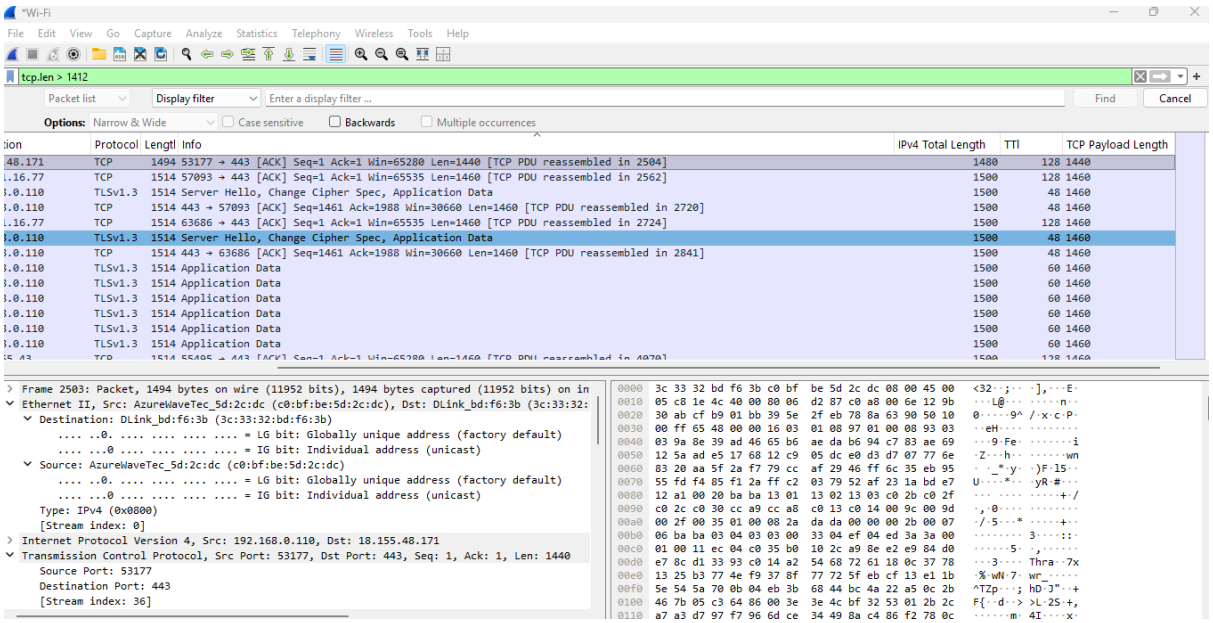


17. Identify packets containing TCP options.



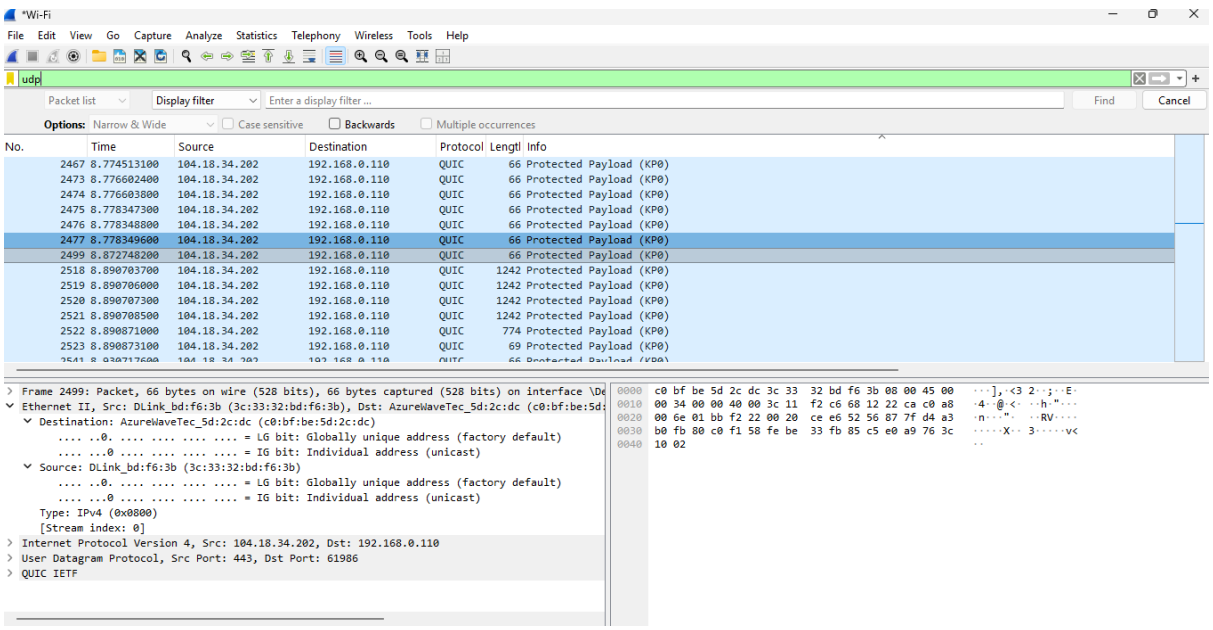


18. Find the TCP packet with the maximum payload length.



Task 5: UDP header inspection

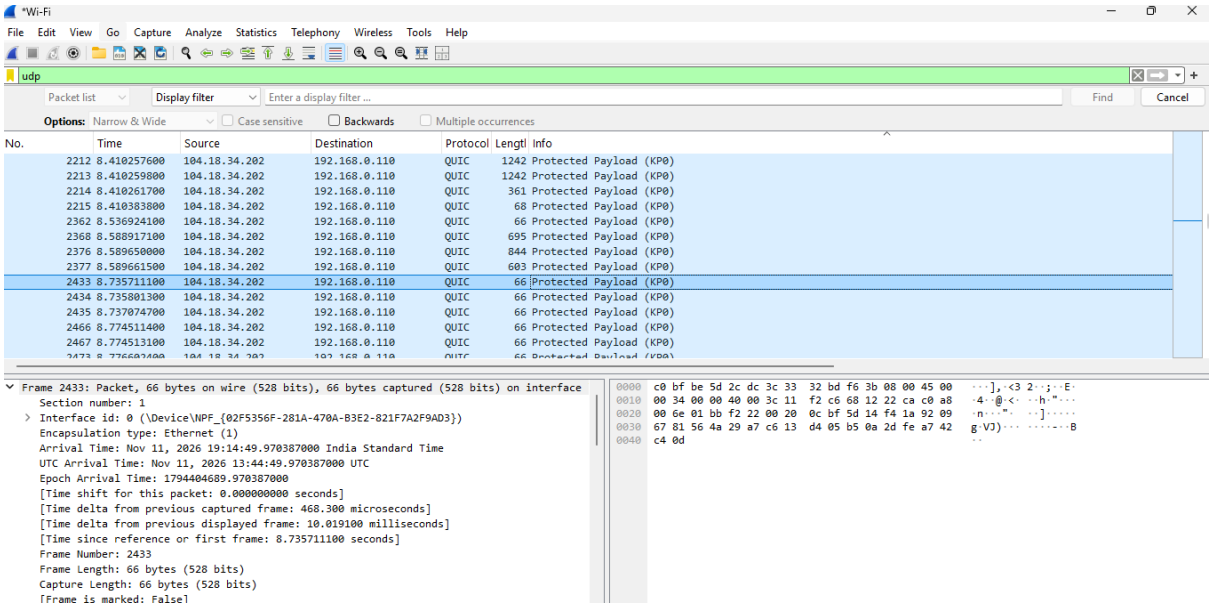
19. Apply the filter udp.

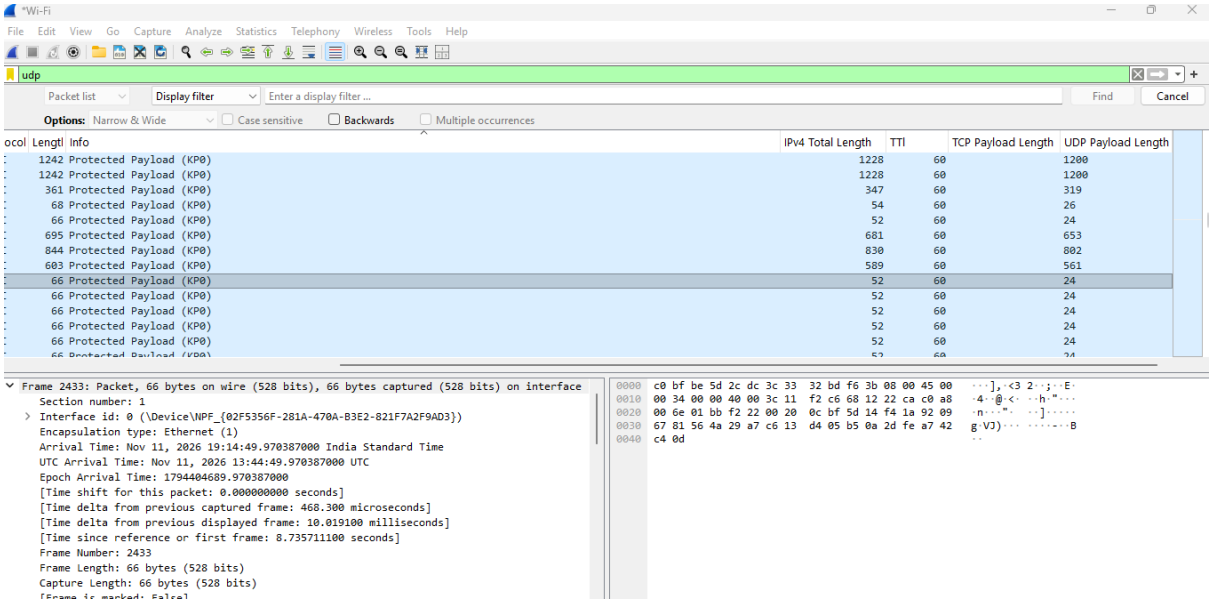


20. Select five UDP packets and record:

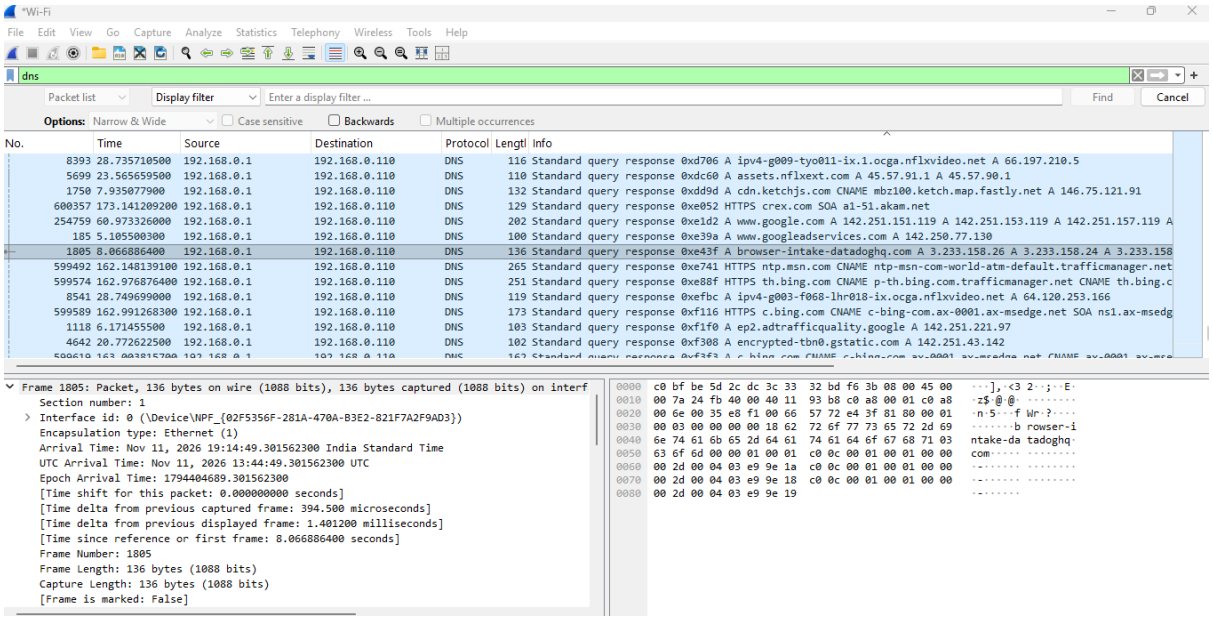
Source port	destination port	udp length	payload length
443	61896	810	802
256	43000	645	636
440	61892	807	798
441	62890	900	896
500	70000	976	965

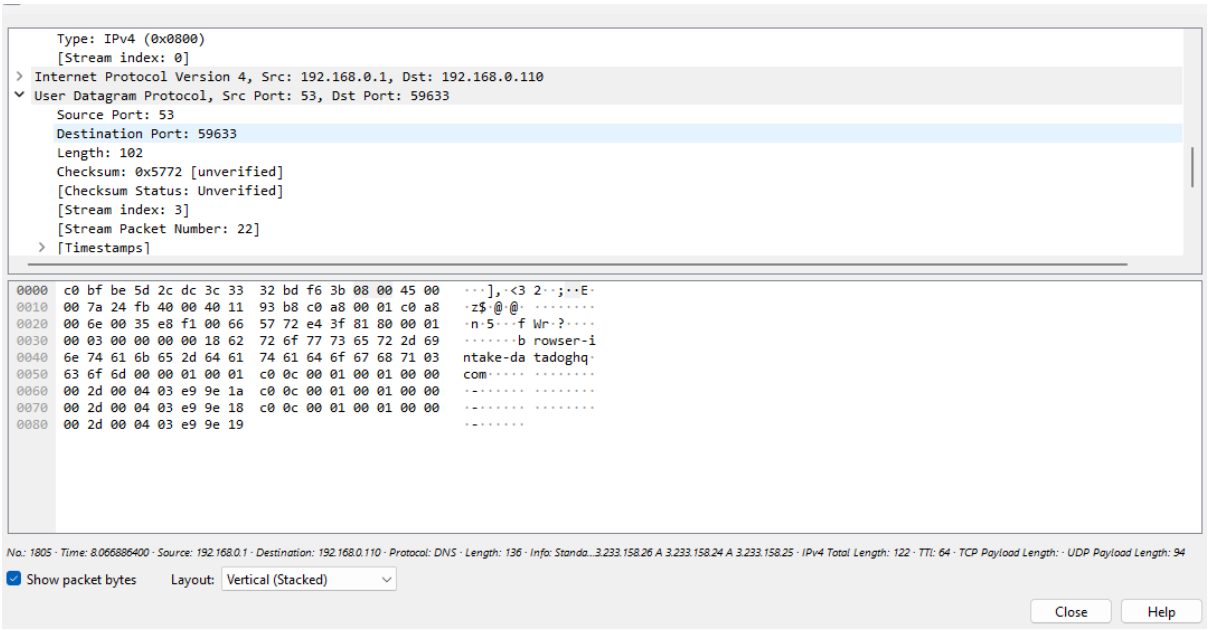
21. Identify the UDP packet with the maximum payload length.





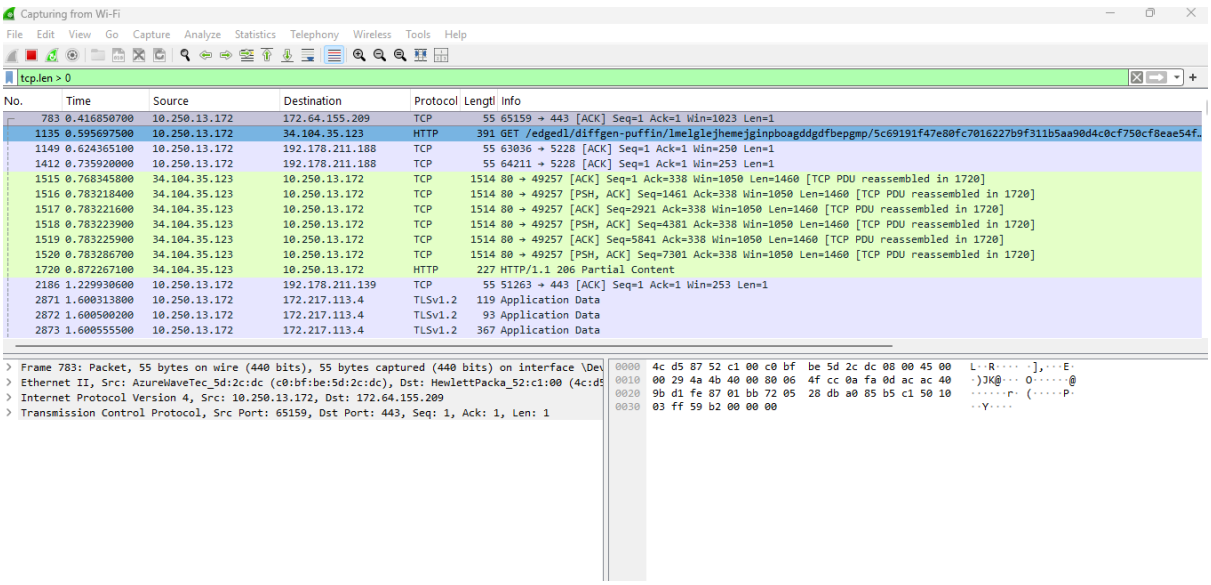
22.Find the UDP packets carrying DNS traffic using the filter dns.

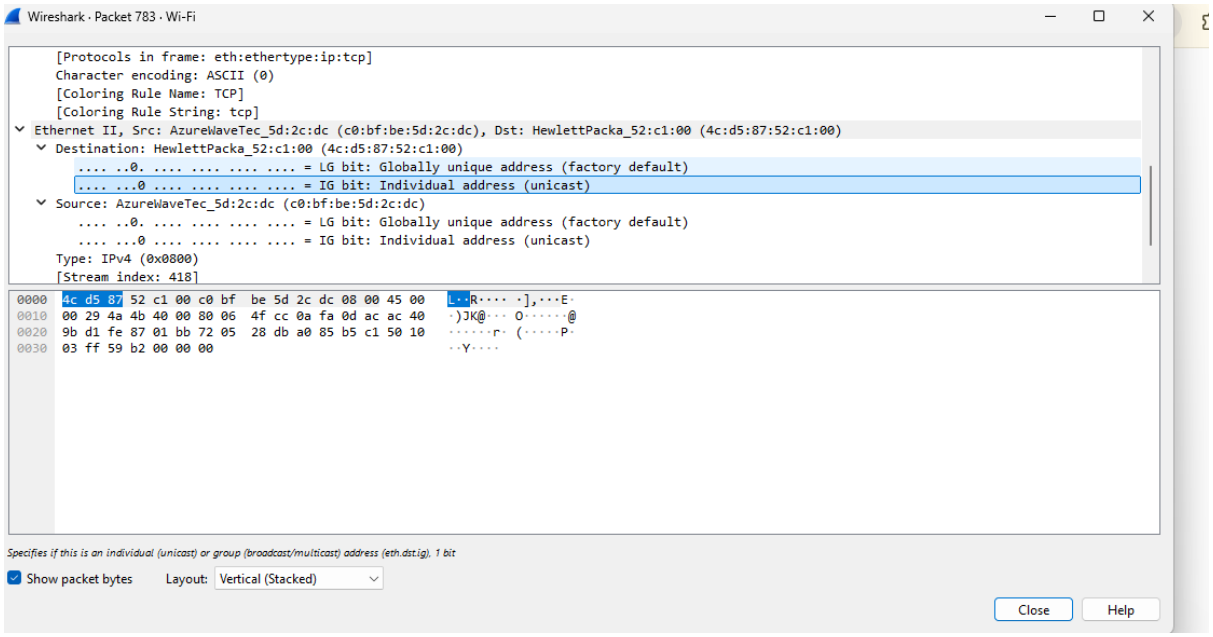


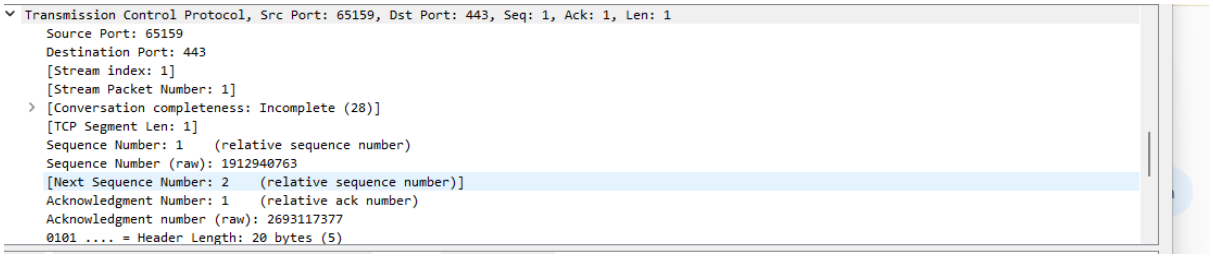


Task 6: Encapsulation and overhead calculation

23. Select one TCP packet containing application data. Record the Ethernet, IPv4 and TCP header lengths.







TCP Segment Len:1

Selected Packet Number:783

Ethernet Header Length:14 bytes

IPv4 Header Length:20 bytes

TCP Header Length:20 bytes

24.Calculate the total protocol header overhead for the selected packet.

Ethernet header:14bytes

Ipv4header:20bytes

Tcpheader:20bytes

Total overhead:14+20+20=54bytes

25.Calculate the application payload size.

Detail	Length
Total frame length	55bytes
Ethernet header	14bytes
Ipv4header	20bytes
Tcpheader	20bytes
Totalheaderoverhead	54bytes

$$\text{Application Payload} = \text{Frame Length} - \text{Header Overhead}$$

=55-54=1

1byte

26.Calculate the payload efficiency percentage.

Total Frame Length	55 bytes
Application Payload	1 byte

$$\text{Payload Efficiency} = \frac{\text{Application Payload}}{\text{Total Frame Length}} \times 100$$

1/55*100=1.82%

27. Repeat the same calculations for one UDP packet.

Detail	Value
Frame Number	856
Total Frame Length	100 bytes
Ethernet Header Length	14 bytes
IPv4 Header Length	20 bytes
UDP Header Length	8 bytes
UDP Length	66 bytes
UDP Payload Length	58 bytes

1.Total Protocol Header Overhead

14+20+8=42

2.Application Payload Size

100-42=58

3. Payload Efficiency

58/100=58%

28.Compare TCP and UDP Header Overhead and Payload Efficiency

Parameter	TCP	UDP
Ethernet Header	14 bytes	14 bytes
IPv4 Header	20 bytes	20 bytes
Transport Header	20 bytes	8 bytes
Total Header Overhead	54 bytes	42 bytes
Application Payload	1 byte	58 bytes

Total Frame Length	55 bytes	100 bytes
Payload Efficiency	1.82%	58%

- TCP has a larger header because it provides reliable, connection-oriented communication.
- UDP has a smaller header of only 8 bytes.
- UDP has lower protocol overhead than TCP.
- In our examples, UDP has higher payload efficiency.